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Senolytic therapy increases replicative capacity by eliminating senescent endothelial cells

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ABSTRACT

Aging is the greatest risk factor for cardiovascular diseases (CVD) and is characterized by inflammation, oxidative stress, and cellular senescence. Cellular senescence is a state of persistent cell cycle arrest triggered by stressors such as DNA damage and telomere attrition. Senescent endothelial cells (ECs) can impair vascular function and promote inflammation, thereby contributing to CVD progression. Senolytics, a class of drugs that selectively eliminate senescent cells, have been shown to remove senescent ECs, but their effects on the replicative capacity and genomic integrity of aged, endothelial cultures remain unclear. In this study, we treated replicative senescent human umbilical vein endothelial cells (HUVECs), used as a model of aged ECs, with Talabostat (10 μM), a novel senolytic, and Navitoclax, 1.0 μM). We hypothesized that senescent cell clearance would be associated with increased proliferative activity and reduced markers of senescence, DNA damage, and telomere dysfunction. Both compounds effectively reduced senescent cell burden, which was accompanied by an increase in cumulative population doublings. Talabostat treatment led to reductions in mean telomere length, DNA damage markers (53BP1), and telomere dysfunction-induced foci (TIFs), suggesting a possible link between increased proliferation and replication-associated telomere attrition. In contrast, Navitoclax treatment increased mitochondrial reactive oxygen species (mtROS) and maintained levels of DNA damage and telomere dysfunction comparable to the vehicle group. Overall, these findings indicate that both Navitoclax and Talabostat can reduce senescent EC burden and promote proliferation in aged EC cultures. Talabostat appears more effective, as it is associated with lower oxidative stress and improved genomic integrity. These results provide insight into how distinct senolytics differentially influence aging-related phenotypes in endothelial cells.

1. Introduction

Aging is the greatest risk factor for cardiovascular diseases (CVD), a leading cause of morbidity and mortality worldwide (Cheng et al., 2023). Advancing age results in increased production of reactive oxygen species (ROS) and heightened inflammation, both of which contribute to arterial dysfunction and the progression of CVD (Schnabel et al., 2008; Donato et al., 2007; Donato et al., 2018). Among the vascular cell types involved, endothelial cells (ECs), which line the inner surface of blood vessels, play a central role in maintaining vascular health by regulating blood flow, coagulation, and immune responses (Bloom et al., 2023a). With age, the endothelium undergoes structural and functional decline, driven in part by oxidative stress and chronic inflammation. These stressors can promote the induction and accumulation of senescent ECs, cells that have persistently exited the cell cycle in response to damage. Senescent ECs impair endothelial function, promote inflammation, and limit the ability of the endothelium to regenerate following injury. Because endothelial regeneration relies on the proliferative expansion of existing ECs, an increased burden of senescent cells may restrict this capacity, further contributing to vascular aging and CVD pathogenesis

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(Donato et al., 2018; Bloom et al., 2023a).

Hayflick and Moorhead first described cellular senescence in the 1960s, noting that normal human fibroblasts undergo a finite number of divisions before entering permanent growth arrest (Hayflick, 1965). This process is driven in part by progressive telomere shortening and genomic instability (De Lange, 2009). Telomeres, repetitive TTAGGG sequences bound by shelterin proteins, protect chromosome ends but shorten with each division, eventually failing to maintain their t-loop structure (De Lange, 2009). When telomeres become critically short, they can no longer maintain their protective t-loop structure or retain shelterin components. As a result, telomeres are recognized as DNA damage, activating the DNA damage response and p53-p21 signaling, which induces senescence (Fagagna et al., 2003; Von Zglinicki et al., 2005). While senescence has beneficial roles in development and tumor suppression (Sager, 1991; Campisi, 2001; Wilkinson and Hardman, 2020), its accumulation contributes to aging and disease by promoting chronic inflammation through the Senescence-Associated Secretory Phenotype (SASP), a mix of inflammatory cytokines, chemokines, growth factors, and ROS (Campisi, 2005; Hornsby, 2002; Coppé et al., 2010). These factors can induce further senescence in neighboring cells, amplifying tissue dysfunction (Lopes-Paciencia et al., 2019; Barinda et al., 2020). In the cardiovascular system, senescent ECs accumulate with age and contribute to pathologies such as atherosclerosis, coronary artery disease, and heart failure (Bloom et al., 2023a). Senescent ECs cannot divide and thus impair endothelial regeneration, which is mediated by mitotic cell division (McDonald et al., 2018). As such, increased senescence burden may underlie diminished vascular repair capacity with age. Whether reducing the burden of senescent ECs can restore replicative capacity and reduce hallmarks of endothelial aging, such as inflammation, ROS, DNA damage, and telomere dysfunction, remains unresolved.

In recent years, there has been a growing interest in targeting senescent cells to combat age-related diseases (Borghesan et al., 2020; Gorgoulis et al., 2019). Numerous studies have shown that small molecules known as senolytics can selectively eliminate senescent cells and improve a range of age-related conditions (Borghesan et al., 2020; Nambiar et al., 2023; Nogueira-Recalde et al., 2019; Saccon et al., 2021; Bloom et al., 2023; Xu et al., 2018; Zhu et al., 2015). These agents have been shown to ameliorate chronic pathologies such as atherosclerosis, liver fibrosis, and impaired stem cell function, contributing to increased health span in preclinical models (Xu et al., 2018; Robbins et al., 2021; Moncsek et al., 2018; Schafer et al., 2017; Ogrodnik et al., 2017). Senolytics are now under investigation in clinical trials to assess their therapeutic potential in humans (Chaib et al., 2022; Power et al., 2023; Yousefzadeh et al., 2018). However, in the context of vascular aging, it remains unclear whether senolytic treatment can enhance the replicative capacity or genomic integrity of aged, non-senescent ECs that persist in a senescent-rich environment.

In this study, we used Navitoclax (ABT-263), a BCL-2 inhibitor, and a novel senolytic compound Talabostat to determine whether senolytic-mediated clearance of senescent cells is associated with enhanced replicative capacity in aged EC cultures. Because telomere shortening occurs heterogeneously across chromosomes and cells, only a subset of aged ECs may reach critically short telomeres and enter senescence, while others retain sufficient telomere length and the potential to divide, particularly if the surrounding senescent burden is reduced. Additionally, the SASP has been proposed to induce senescence in neighboring cells, and its reduction through senescent cell clearance may limit senescence induction in the adjacent non-senescent ECs. Therefore, we focused on whether senolytic treatment would reduce DNA damage markers and telomere dysfunction while enhancing the proliferative potential of aged endothelial cultures following senescent cell clearance. Our findings suggest that senolytic treatment is associated with enhanced replicative capacity and may contribute to improved genomic integrity in aged EC cultures, offering insights into potential strategies to restore endothelial regenerative function with age.

2. Methods

2.1. Cell culture

Human Umbilical Vein Endothelial Cells (HUVEC, Single donor, Lonza C2517A) were cultured on T-72 flasks coated with 1 % fibronectin mimetic at a density of 1.4×10^6 cells per plate. The cells were maintained at 37 °C with 5 % CO^2, using Endothelial Cell Growth Medium-2, EGM-2 (CC-3162, Lonza). Passaging was performed every 6–7 days regardless of confluency. For the experiment, young cells were passaged up to the 4th generation, while replicative senescence in the “old” group was identified by monitoring cumulative population doublings (CPD) over time. Cells were seeded at day 0 and passaged under standard conditions, with CPD recorded at each passage. On day 43 after initial seeding, the CPD curve began to plateau, indicating a noticeable slowdown in proliferation and the onset of senescence (Fig. S1). This point was chosen for senolytic treatment to target cells at the senescence threshold rather than earlier proliferative stages and each assay was performed at defined time points (Fig. S3). During passaging, a 10 μL aliquot of cell suspension stained with 0.1 % trypan blue and used for counting. Between 0.3×10^6 and 0.5×10^6 cells were plated onto fresh 35 mm plates for experimental measurements. The CPD values were calculated using the following formula (Donders et al., 2018):

$$3.32 \times [(\log(\text{harvested cells})) - (\text{Log}(\text{seeded cells}))] + \text{previous PD}$$

Two independent HUVEC batches were used, each including a vehicle group (7–8 replicates) and treatment groups (4 replicates). To combine data from these batches and eliminate inter-batch variability, we calculated the “net number of doublings,” defined as the increase in PD between two time points. This metric was expressed in arbitrary units (a.u.), as absolute values were normalized to the corresponding Old (vehicle) group, enabling direct comparison across independent experiments and cell batches. In this context, “a.u.” refers to normalized PD change rather than an absolute physical unit. Replicative senescent HUVECs underwent one of three treatments for 24 h: 1) DMSO as vehicle control, 2) 1.0 μM of Navitoclax (ABT-263), or 3) 10 μM of a novel senolytic compound, Talabostats. Following treatment, the medium was replaced with fresh EGM-2, and experiments were conducted within a specified timeline described in the corresponding assay subsections.

2.2. Senescence-associated β-galactosidase (SA-βgal)

To verify cell senescence, X-Gal staining was performed 48 h post-treatment using the Senescence β-galactosidase Staining Kit (Cat No. 9860, Cell Signaling Technology) according to the manufacturer's instructions. Cells were initially fixed in a 1× fixative solution for 15 min at room temperature. After washing with 1× PBS, 1 ml of β-galactosidase staining solution was added to each 35 mm well. The cells were then incubated overnight at 37 °C in a dry incubator without CO^2. Images were captured using an EVOS XL core Amex 1200 microscope at 20× magnification, counting nine sites per well for each experimental group.

2.3. Bromodeoxyuridine (BrdU) cell proliferation assay

The BrdU Cell Proliferation ELISA Kit (Cat No. ab126556, Abcam) was utilized to assess cell proliferation at 5, 29, and 70 days post-treatment according to the manufacturer's protocol. Briefly, cells were incubated with BrdU labeling reagent for 12 h to allow incorporation into newly synthesized DNA. Following fixation, cells were incubated with an anti-BrdU antibody conjugated to horseradish peroxidase (HRP). After washing, the HRP substrate solution was added, and chemiluminescence was measured at 450 nm using a microplate reader. Optical density (OD) values were normalized to the Old (vehicle) within each experiment and reported as a measure of relative DNA synthesis.

2.4. Quantitative real-time polymerase chain reaction (qRT-PCR)

Total RNA was extracted from HUVECs 5 days post-treatment using the RNeasy Mini Kit (Cat. No. 74104, Qiagen) following the manufacturer's protocol. The quality of the RNA was assessed using a Nanodrop spectrophotometer, with an absorbance ratio (260/280) above 1.8, indicating sufficient quality. mRNA was then reverse transcribed into cDNA utilizing the QuantiTect Rev. Transcription Kit (Cat. No. 205311, Qiagen) as per the recommended guidelines. Quantitative PCR was carried out using SsoFast EvaGreen Supermixes (Bio-Rad) on the Bio-Rad CFX™ Real-Time system. The 18s gene served as the normalization control, with data analysis performed using the $2^{-\Delta\Delta Ct}$ method (Livak and Schmittgen, 2001). Fold changes in gene expression were expressed relative to the Old (vehicle) group, which was set to 1. Primer sequences are detailed in Table S1.

2.5. Immunofluorescence fluorescent in situ hybridization (IF-FISH)

Ten days post-treatment, IF-FISH was conducted on the HUVECs following the established protocols (Bloom et al., 2022). Initially, cells were fixed in 4 % PFA and then immersed in cold 100 % methanol at −20 °C for 15 min. This was followed by rehydration in 1× PBS at room temperature for 5 min. The cells were then placed in a blocking solution (comprising 1 mg/ml BSA, 3 % goat serum, 0.1 % Triton X-100, and 1 mM EDTA in 1× PBS) for 30 min at room temperature. Subsequently, cells were incubated in the blocking solution with a 1:500 dilution of the 53BP1 antibody (Cat No. NB 100-0304, Rabbit, Novus Biologicals) for 1 h at room temperature. After three washes of 5 min each in 1× PBS, the samples were incubated for 1 h at room temperature with a 1:500 dilution of Alexa Fluor 555 in the blocking solution (Cat No. A-21429, Goat anti-Rabbit, Invitrogen).

Cells then underwent dehydration in 70 %, 95 %, and 100 % ethanol for 5 min each. The ethanol was then removed, and the samples were briefly air-dried before exposure to a hybridization solution. This solution contained 2 % Tris HCl, 60 % formamide, 5 % blocking reagent from a 10 % Roche Stock (dissolved in maleic acid buffer with 100 mM maleic acid, 150 mM NaCl, pH 7.4, Millipore Sigma), and a 1:200 dilution of a Tel probe (Integrated DNA Technologies, 5'-Alex488N/CC CTA ACC CTA ACC CTA A, purified via high-performance liquid chromatography) in deionized H2O (diH2O). The samples were incubated at 60 °C for 10 min and then at 85 °C for 10 min. Subsequently, cells were gently agitated in a washing solution of 2× saline sodium citrate (SSC) and 0.1 % Tween 20 at 60 °C for 10 min, repeated twice. The cells were then washed in 2× SSC, 1× SSC, and diH2O, each for 10 min at room temperature. After brief air-drying, samples were mounted with DAPI Fluoromount G (Cat No. 102092-102, VWR), weighed with a 1 kg weight for 5 min, and stored in a dark container until imaging.

2.6. Imaging and analysis

Samples were imaged using an Olympus Fluoview FV1000 confocal microscope at 100× magnification with uniform acquisition settings applied across all samples. *Z*-slices, each 1 μm thick, were taken throughout the entire nucleus, and an identical number of z-slices were stacked for each cell during analysis. The images were processed using CellProfiler (https://cellprofiler.org), and telomere length was assessed indirectly by measuring telomere fluorescence intensity. For each nucleus, the integrated fluorescence intensity of all telomere spots was calculated, and values were normalized to the mean intensity of the Old (vehicle) group. Data are expressed in arbitrary units (a.u.), where "a.u." refers to normalized fluorescence intensity values (relative to vehicle = 1) rather than absolute intensity units. Quantification of a total of 53BP1 foci and 53BP1 foci colocalizing with telomere signals (telomere dysfunction-induced foci [TIF]) was also performed using CellProfiler. At least 50 cells were measured per data point. In total, we analyzed 301 cells from the Young (p4) group, 424 cells from the vehicle-treated group, 352 cells from the Talabostat-treated group, and 312 cells from the Navitoclax-treated group.

2.7. MtROS measurement

Mitochondrial reactive oxygen species (mtROS) were measured 5 days post-treatments using the fluorescent probe MitoSOX (Cat No. M36008, Invitrogen) according to the manufacturer's protocol. In brief, HUVECs were incubated with 5 μM MitoSOX in culture medium for 10 min. The cells were then washed three times with prewarmed 1× PBS, mounted using DAPI Fluoromount G (Cat No. 102092–102, VWR), and visualized under an Olympus Fluoview FV1000 confocal microscope at 20× magnification. Fluorescence intensity per cell was quantified using CellProfiler, and mtROS levels were normalized to the mean value of the Old (vehicle) group. Results are presented in arbitrary units (a.u.), where "a.u." refers to normalized fluorescence intensity.

2.8. Statistical analysis

Individual data points represent the mean value of all cells analyzed within each experimental replicate. Statistical analysis was performed using GraphPad Prism 8.4.3. Group differences were assessed via repeated measures one-way analysis of variance (ANOVA) with LSD post hoc tests. Frequency distributions of telomere length were analyzed using the Kruskal-Wallis non-parametric test with uncorrected Dunn's post hoc test. Significance was determined at $p < 0.05$. Data are presented as the mean ± SEM.

3. Results

3.1. Efficacy of senolytics to remove senescent endothelial cells

To evaluate the effectiveness of Navitoclax and Talabostat in reducing senescence burden, we assessed SA β-gal staining in replicative senescent ECs and young controls. As expected, the vehicle-treated senescent ECs exhibited a significantly higher proportion of SA–β–gal–positive cells compared to young ECs (Fig. 1A, B; $p < 0.001$). Forty-eight hours after senolytic treatment, both Navitoclax and Talabostat significantly reduced the percentage of SA–β–gal–positive cells relative to vehicle controls (both $p < 0.001$). Consistent with this, mRNA expression of senescence markers, including the cyclin-dependent kinase inhibitors p16 and p21, and the tumor suppressor p53, was elevated in the vehicle group compared to young ECs (Fig. 1C; all $p < 0.001$). Five days post-treatment, both Navitoclax and Talabostat significantly reduced the expression of all senescence markers relative to vehicle ($p < 0.001$).

Senescent ECs are known to secrete a range of pro-inflammatory and tissue-remodeling factors collectively referred to as the Senescence-Associated Secretory Phenotype (SASP), which can induce senescence in surrounding cells, causing vascular dysfunction (Lopes-Paciencia et al., 2019; Barinda et al., 2020; Stojanović et al., 2020). To examine whether senolytic treatment affects SASP expression, we analyzed mRNA levels of canonical SASP components in cell lysates. The SASP factors analyzed included IL-6, IL-8, MCP-1, PAI-1, TGF-β, CXCL1, CXCL2, CXCL10, and MMP-3, all of which were significantly elevated in the vehicle group compared to young ECs (Fig. 1D; all $p < 0.05$). Treatment with either senolytic compound reduced the expression of all evaluated SASP factors relative to vehicle. These findings suggest that both Navitoclax and Talabostat treatment is effective at reducing the burden of replicative senescent ECs and mitigating the harmful secretory phenotype associated with these cells.

3.2. Senolytics reduce the suppressive impact of senescent cells on the proliferation of aged endothelial cells

We aimed to assess whether senolytic treatment was associated with

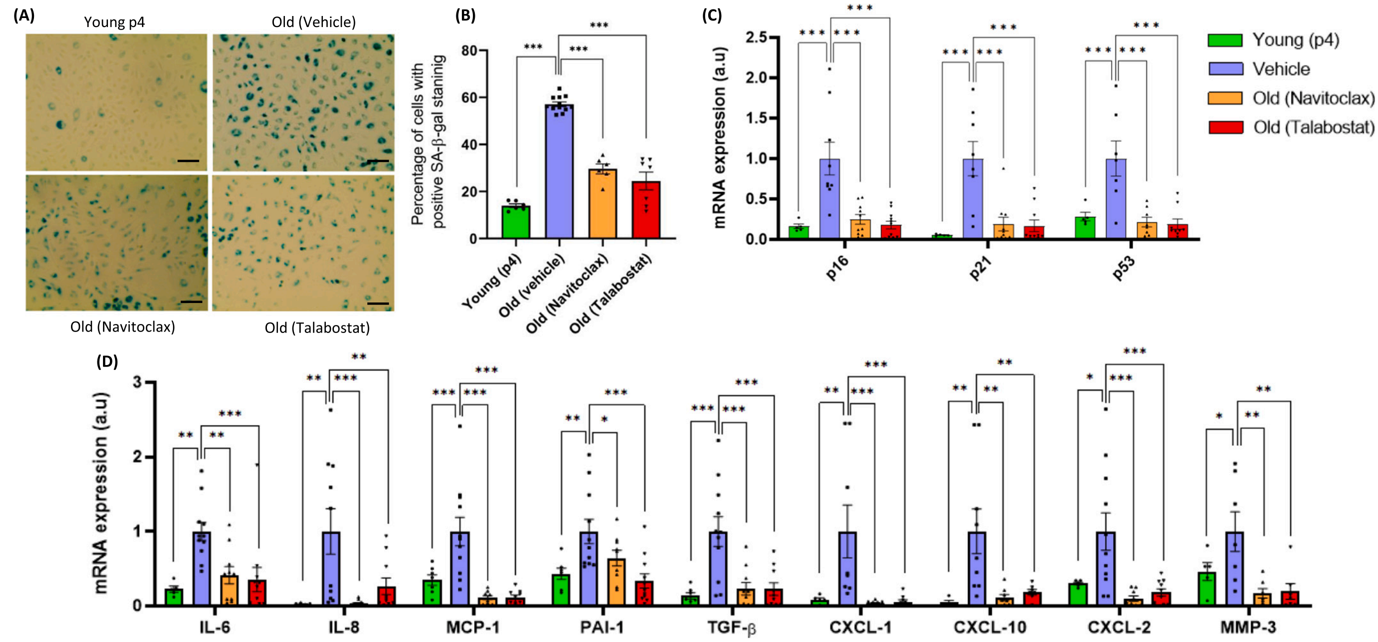

Fig. 1. Senolytics remove replicative senescent Human Umbilical Vein Endothelial Cells (HUVECs). A. Representative images of senescence-associated beta-galactosidase (SA-β-gal) staining 48 h after treatment and B. quantification of SA-βgal-positive HUVECs. C. Relative mRNA expression of senescence markers (p16, p21, and p53) and D. Senescence-associated secretory phenotype (SASP) factors mRNA expression 5 days post-treatment. For all bar graphs, "a.u." (arbitrary units) refers to normalized fold change values rather than absolute expression units. Images were taken using 20× magnification. Scale bar 20 μm. $N = 5$–12 experimental replicates. * = $p < 0.05$, ** = $p < 0.005$, *** = $p < 0.001$, a.u.—arbitrary units. One-way ANOVA with LSD post-hoc test was used to determine group differences. Data are mean ± SEM.

improved replicative capacity in aged EC cultures following the removal of replicative senescent cells. Cumulative population doublings (CPD) were monitored for 5 days before and up to 29 days after treatment with either Navitoclax, Talabostat, or vehicle control. During the 5-day pre- and post-treatment period, CPD values showed slight variation but did not differ significantly between groups (Fig. S2). However, by day 29 post-treatment, the PD curves diverged, with both senolytic-treated groups exhibiting continued increases in CPD, whereas the vehicle group plateaued, showing minimal further proliferation. Despite some batch variability, trends were consistent across replicates. To quantify changes in proliferation, we compared PD values from pre- to post-treatment at day 5 and from day 5 to day 29 post-treatment. Both Navitoclax and Talabostat significantly increased the net number of doublings compared to vehicle (Fig. 2A; $p < 0.001$). Similarly, BrdU

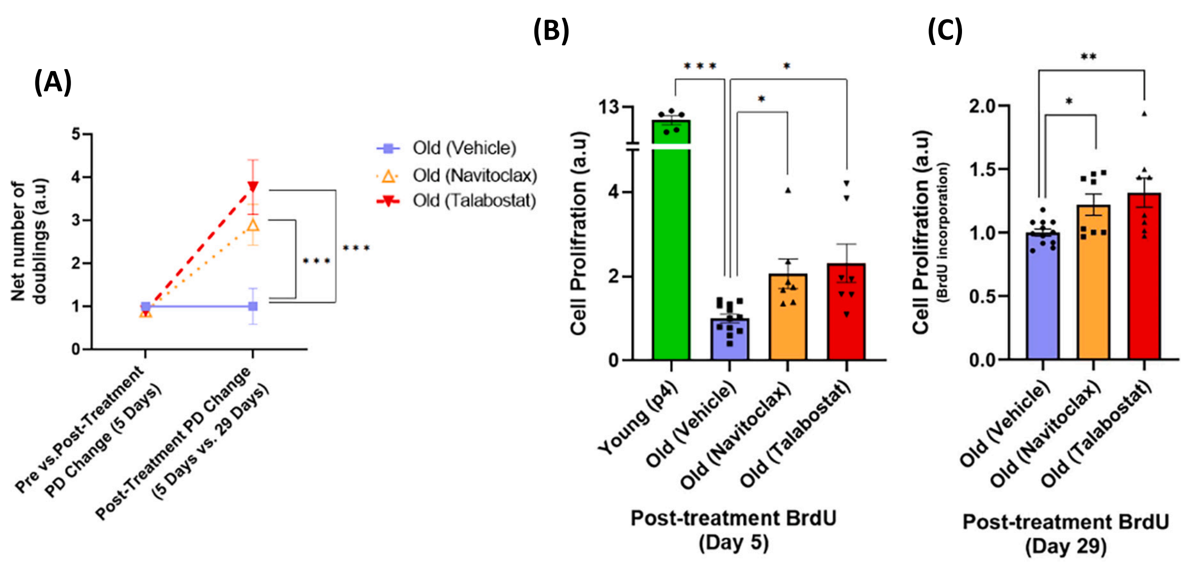

Fig. 2. Senolytics restore the proliferative capability of Human Umbilical Vein Endothelial Cells (HUVECs). A. Net number of doublings between two time intervals: day −5 to day 5 (pre- vs. post-treatment) and day 5 to day 29 post-treatment. "Net number of doublings" was calculated from population doubling (PD) values using the formula 3.32 × [log(harvested cells) − log(seeded cells)] + previous PD. Values are expressed in arbitrary units (a.u.), where a.u. refers to normalized PD change relative to the vehicle group within each independent experiment, allowing comparison across HUVEC batches. B. Bromodeoxyuridine (BrdU) incorporation in young, vehicle-treated old, Navitoclax–treated, and Talabostat–treated endothelial cells at 5 days post-treatment and C. 29 days post-treatment. BrdU incorporation was measured by ELISA (Optical density (OD), 450 nm) as an indicator of DNA synthesis and relative cell proliferation. N = 5–15 experimental replicates. * = $p < 0.05$, ** = $p < 0.005$, *** = $p < 0.001$, a.u.—arbitrary units. One-way ANOVA with LSD post-hoc test was used to determine group differences. Data are mean ± SEM.

incorporation assays showed that the vehicle group had significantly lower proliferation than the young control at 5 days post-treatment (Fig. 2B; $p < 0.001$). In contrast, both senolytic-treated groups exhibited approximately double the proliferation rate of the vehicle group ($p = 0.044$ and $p = 0.015$ for Navitoclax and Talabostat, respectively). Although proliferation declined in all groups by day 29, it remained significantly higher in senolytic-treated cultures than in the vehicle group ($p < 0.05$). By day 70, proliferation ceased in all conditions (data not shown). These findings suggest that clearance of senescent ECs by Navitoclax or Talabostat may alleviate barriers to proliferation in aged EC cultures, prevent senescence induction on aged ECs, and enable the remaining ECs to regain their proliferative capability and resume proliferation.

3.3. Effect of senolytics on aged endothelial cells, DNA damage, and telomere dysfunction

Senescent cells exhibit persistent DNA damage foci, which contribute to their dysfunctional phenotype and are associated with pro-inflammatory signaling (Galbiati et al., 2017; Rodier et al., 2011). To evaluate the effects of senolytics on DNA damage, we assessed 53BP1 foci, a marker of double-stranded DNA breaks, in replicative senescent ECs 10 days post-treatment. The vehicle group exhibited a significantly higher percentage of ECs with one or more 53BP1 foci compared to young ECs (Fig. 3A, B; $p < 0.001$). Likewise, the number of 53BP1 foci per EC was elevated in vehicle-treated cells relative to the young group (Fig. 3C; $p < 0.001$). Talabostat treatment significantly reduced both the percentage of ECs with DNA damage and the number of 53BP1 foci per cell compared to the vehicle group (Fig. 3A–C; $p < 0.001$ and $p = 0.003$, respectively), restoring levels comparable to those observed in young ECs ($p = 0.332$ and $p = 0.383$). In contrast, Navitoclax treatment did not significantly change either the percentage of 53BP1-positive cells or the number of foci per cell compared to the vehicle group. These findings suggest that Talabostat is more effective than Navitoclax in reducing markers of DNA damage in aged EC cultures.

In addition to general genomic damage, telomeres are particularly vulnerable to DNA damage, and telomere dysfunction is a hallmark of replicative senescence (Bloom et al., 2023b). To assess telomere-specific damage, we measured telomere-induced foci (TIFs), DNA damage signals that colocalize with telomeres. Vehicle-treated replicative senescent ECs displayed a significantly higher percentage of cells with one or more TIFs and a greater number of TIFs per cell compared to young controls (Fig. 4A–C; $p < 0.001$ and $p = 0.002$, respectively). Talabostat treatment significantly reduced both the percentage of TIF-positive ECs and the number of TIFs per cell relative to vehicle ($p = 0.001$ and $p = 0.007$), bringing both measures to levels not different from those observed in young ECs ($p = 0.310$ and $p = 0.406$). In contrast, Navitoclax treatment did not result in significant changes in TIF frequency or number compared to the vehicle group. Together, these results indicate that Talabostat is associated with a marked reduction in both total and telomere-specific DNA damage in replicative senescent ECs. In contrast, Navitoclax did not show similar efficacy under the same conditions, suggesting possible differences in the mechanisms or how these senolytic agents influence genomic stability in aged endothelial cultures.

3.4. Impact of senolytics on telomere dynamics in replicative senescent endothelial cells

To assess the effects of senolytics on telomere dynamics, we measured telomere fluorescence intensity, a measurement for telomere length, in replicative senescent ECs 10 days post-treatment. As expected, the mean telomere length in the vehicle group was significantly shorter than in the young ECs (Fig. 5A; $p < 0.001$). Treatment with Talabostat led to a further reduction in mean telomere length compared to vehicle ($p < 0.001$), which can reflect increased proliferation in non-senescent ECs, where telomere attrition occurs with each cell division. This interpretation is supported by telomere length distribution analysis, which showed a higher frequency of short telomeres in the Talabostat group relative to vehicle (Fig. 5B; $p = 0.032$). Despite this shift toward shorter telomeres, Talabostat-treated ECs continued to proliferate through day 29 post-treatment (Fig. 2C), suggesting that telomeres, while shorter, were not critically short and remained compatible with cell division.

In contrast, Navitoclax treatment increased proliferation but did not significantly alter mean telomere length or its distribution compared to the vehicle group ($p = 0.801$ and 0.886, respectively), suggesting that enhanced proliferation occurred independently of detectable telomere shortening. Overall, these findings suggest that Talabostat may promote

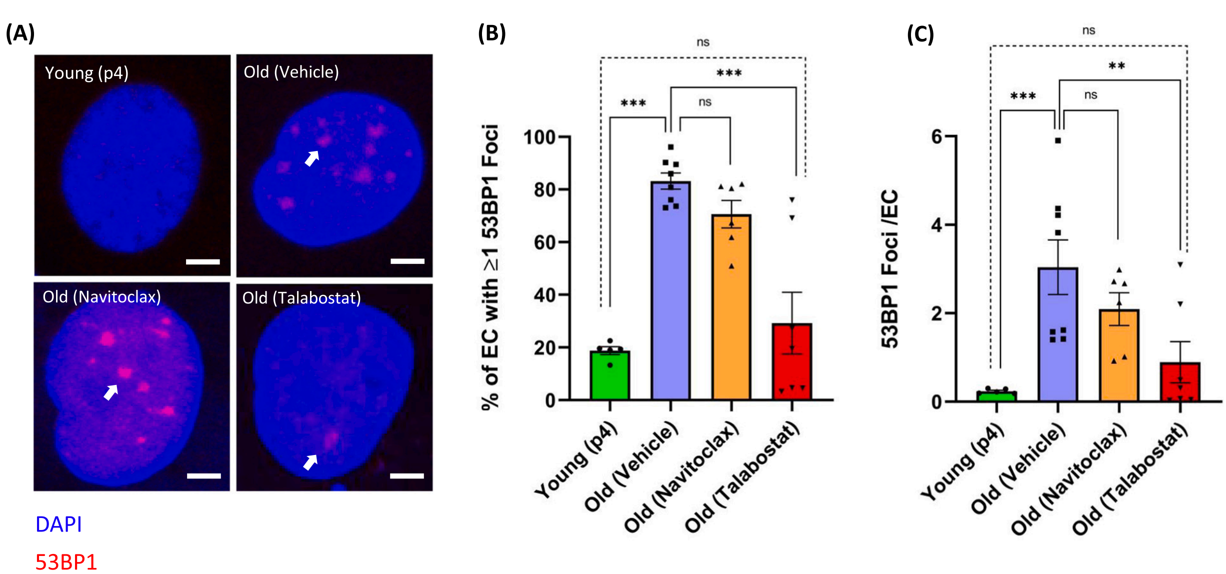

Fig. 3. Effect of senolytics on Human Umbilical Vein Endothelial Cells (HUVECs) DNA damage. A. Representative images of immunofluorescence (IF) staining for the 53BP1, a double-strand DNA break marker, as shown with the white arrow, in the Young or replicative senescent HUVECs treated with vehicle, Navitoclax or Talabostat B. The percentage of HUVECs with one or more 53BP1 foci. C. The number of 53BP1 foci per HUVEC. One-way ANOVA with LSD post-hoc test was used to determine group differences. Images were taken using 100× magnification. N = 5–8 replicates per group, ** = $p < 0.005$, *** = $p < 0.001$. ns = none significant. Scale bars = 5 µm. Data are mean ± SEM.

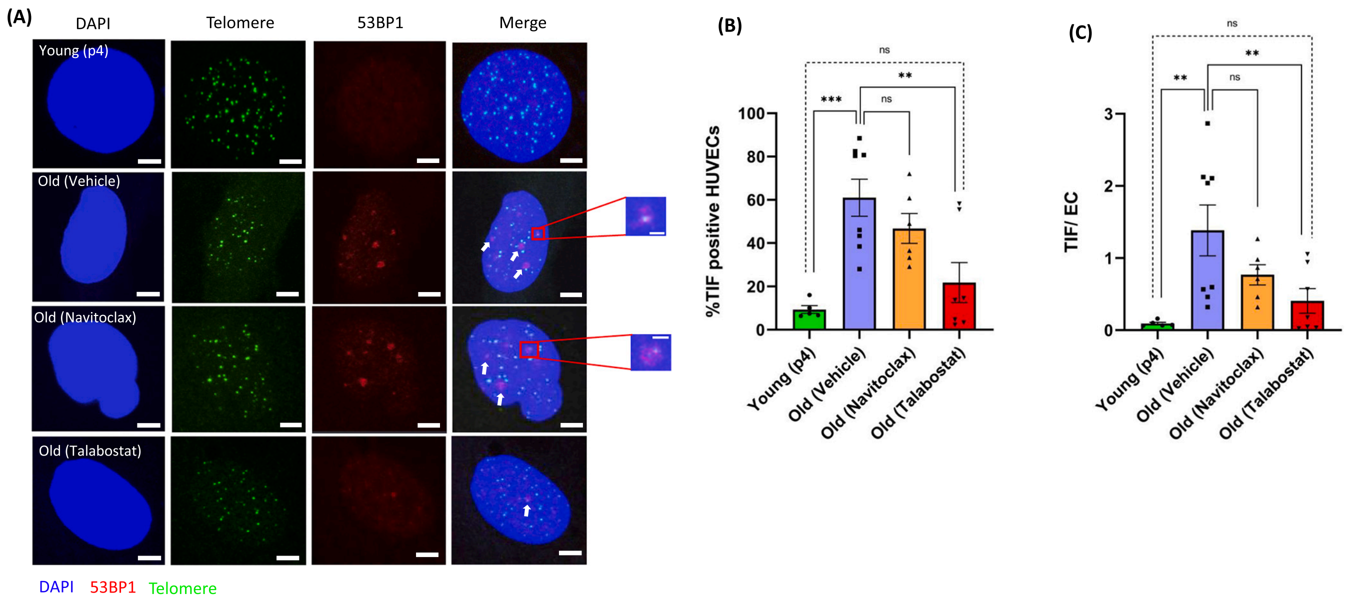

Fig. 4. Effect of senolytics on Human Umbilical Vein Endothelial Cells (HUVECs) Telomere Dysfunction-Induce Foci (TIF). A. Representative images of immunofluorescence fluorescent in situ hybridization (IF-FISH), detecting the DNA damage marker 53BP1 (red) and telomeres (green) in Young (p4) HUVECs, as well as HUVECs treated with vehicle, Navitoclax, and Talabostat. Colocalization of 53BP1 and telomere signals indicate telomere dysfunction-induced foci (TIF). B. Percentage of HUVECs with one or more TIF. C. The number of TIF per endothelial cells (ECs). One-way ANOVA with LSD post-hoc was used to determine group differences. Images were taken using 100× magnification. Scale bar 5 μm and 2 μm. N = 5–8 replicates per group. ns = none significant,** = p < 0.005, *** = p < 0.001. Data are mean ± SEM. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

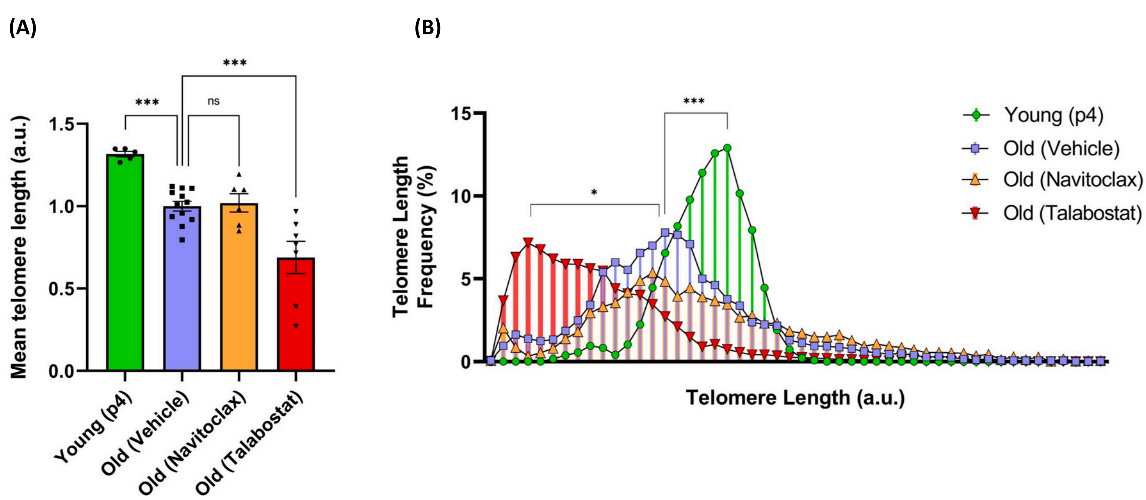

Fig. 5. Effect of the senolytics on Human Umbilical Vein Endothelial Cells (HUVECs) telomere length. A. Mean telomere length was quantified from the fluorescence intensity of immunofluorescence fluorescent in situ hybridization IF-FISH in HUVECs treated with vehicle Navitoclax or Talabostat. Telomere fluorescence intensities were normalized to the mean value of the Old (Vehicle) group and expressed as arbitrary units (a.u.), where "a.u." refers to normalized fluorescence intensity (relative to Old (vehicle) = 1). One-way ANOVA with LSD post-hoc test was used to determine group differences. B. Frequency distribution of telomere lengths. N = 5–12 experimental replicates. * = p < 0.05, *** = p < 0.001, ns = none significant, a.u.—arbitrary units. Kruskal-Wallis non-parametric test with uncorrected Dunn's test post hoc was used to determine group differences. Data are mean ± SEM.

replicative activity in aged endothelial cultures through telomere-dependent mechanisms, while Navitoclax may act via alternative, telomere-independent pathways.

3.5. Impact of senolytics on Mitochondrial Reactive Oxygen Species (mtROS) in replicative senescent endothelial cells

To investigate potential reasons for the different effects of Navitoclax and Talabostat on DNA damage and telomere dysfunction, we evaluated mitochondrial reactive oxygen species (mtROS) production, as oxidative stress is a known driver of genomic instability (Klaunig et al., 2011). Navitoclax is a BCL-2 family inhibitor that induces apoptosis through mitochondrial membrane permeabilization and caspase activation (Pandravada and Sandler, 2021), processes that can elevate mtROS levels as part of the apoptotic cascade (Oh et al., 2021; Lagadinou et al., 2013). Consistent with prior findings, mtROS levels were significantly higher in vehicle-treated replicative senescent ECs than in young ECs (Fig. 6A, B; $p < 0.001$). Navitoclax treatment further increased mtROS compared to vehicle ($p < 0.001$), whereas Talabostat significantly reduced mtROS levels relative to both vehicle and Navitoclax-treated groups ($p = 0.026$ and $p < 0.001$, respectively). These results highlight mechanistic differences between the two senolytics. While

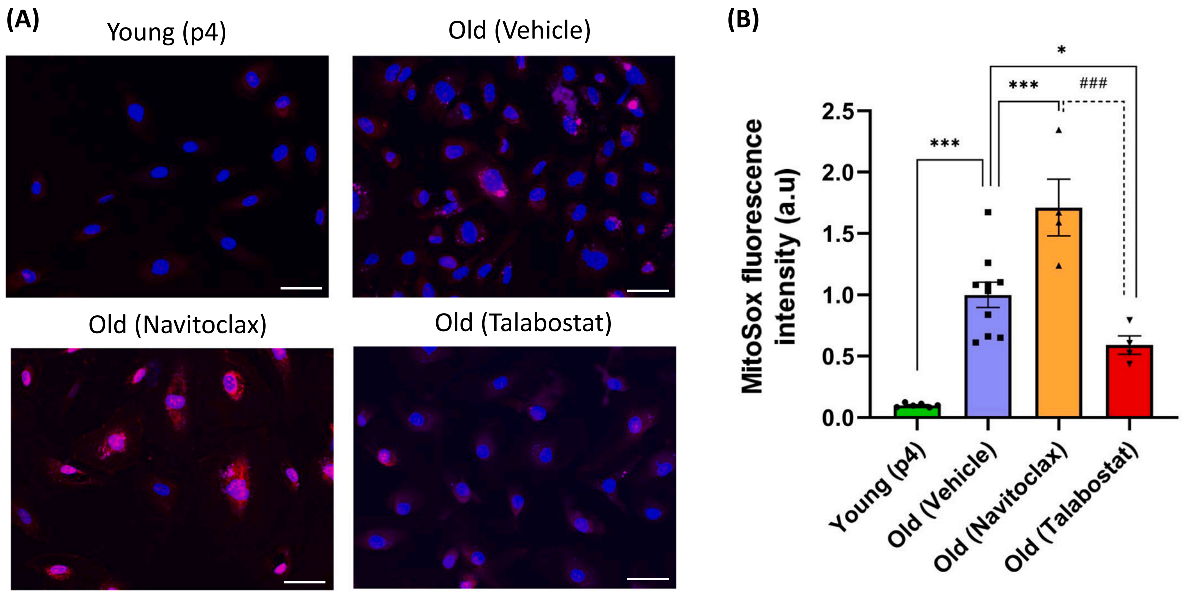

Fig. 6. Effect of senolytics on the Mitochondrial Reactive Oxygen Species (mtROS) in replicative senescent Human Umbilical Vein Endothelial Cells (HUVECs). A. Representative images of mitochondrial superoxide production in HUVECs, assessed using the MitoSOX fluorescent probe. B. Quantitative analysis of mtROS levels based on MitoSOX fluorescence intensity measured with CellProfiler. Values were normalized to the mean intensity of the Old (Vehicle) group and expressed in arbitrary units (a.u.), where "a.u." refers to normalized fluorescence intensity (relative to Old (Vehicle) = 1). The images were taken at 20× magnification, and the scale bar = 20 μm. $N = 4$–10 replicates per group. $* = p < 0.05$, $***/^{\#\#\#} = p < 0.001$, ns = none significant. a.u.—arbitrary units. One-way ANOVA with LSD post-hoc test was used to determine group differences. Data are mean ± SEM.

Navitoclax exacerbates oxidative stress through its pro-apoptotic activity, Talabostat appears to confer a protective effect by reducing mitochondrial ROS in aged ECs. This difference in mtROS modulation may contribute to the reduction in DNA damage and telomere dysfunction observed with Talabostat treatment.

4. Discussion

The key findings of this study are as follows: 1) both Navitoclax and a novel senolytic compound, Talabostat, effectively reduced senescent endothelial cells. 2) This reduction in senescence burden was associated with increased proliferation in the aged endothelial cell population. 3) In Talabostat-treated cultures, increased proliferation was associated with shorter telomere length, reduced DNA damage, and fewer TIFs; and 4) in contrast, Navitoclax treatment led to increased markers of DNA damage and TIFs, along with elevated mtROS. These results support the hypothesis that senolytic treatment enhances the replicative potential of aged endothelial cells and suggest that distinct senolytic agents differentially affect the genomic integrity and telomere dynamics of the remaining cell population. Collectively, our findings highlight a potential mechanistic advantage of Talabostat over Navitoclax.

Cellular senescence contributes to aging-related tissue dysfunction by impairing cell function and promoting chronic inflammation through the secretion of SASP factors (Campisi, 2005; Hornsby, 2002; Coppé et al., 2010). In endothelial cells, senescence is associated with impaired vascular homeostasis, reduced regenerative capacity, and progression of cardiovascular diseases such as atherosclerosis and hypertension (Bloom et al., 2023a; Wang et al., 2024; Han and Kim, 2023; Seals et al., 2011; Molnár et al., 2023). Senolytics, a class of drugs designed to selectively induce cell death in senescent cells, have emerged as promising candidates to mitigate these effects (Borghesan et al., 2020; Gorgoulis et al., 2019; Xu et al., 2018; Zhu et al., 2015). However, most studies have focused on the clearance of senescent cells themselves, with limited attention to how this affects the behavior of remaining cells within aged populations. In this study, we compared a well-characterized BCL-2 inhibitor, Navitoclax (Zhu et al., 2016), with a novel senolytic compound, Talabostat, to evaluate whether the removal of senescent cells is associated with enhanced proliferative potential and improved cellular health in aged endothelial cell cultures. Importantly, our approach extends prior work by evaluating not only senescent cell elimination but also downstream outcomes in aged EC populations, including proliferation, genomic integrity, and telomere dynamics. Given the growing clinical interest in senolytic therapies (Justice et al., 2019; Hickson et al., 2019; Singh et al., 2016), understanding their broader cellular effects, particularly in vascular contexts, is essential.

Navitoclax is a senolytic agent that induces cell death in senescent cells by inhibiting BCL-2 family proteins, and it has demonstrated efficacy in preclinical models across multiple tissues (Zhu et al., 2016). In aged mice, Navitoclax has been shown to clear senescent hematopoietic stem cells and restore regenerative potential, as measured by improved clonogenicity and engraftment (Chang et al., 2016). In the present study, we found that Navitoclax reduced senescent endothelial cell burden and was associated with increased replicative activity in aged EC cultures. These results expand upon earlier reports using HUVECs as a model system, highlighting the potential for cell-type–specific responses to Navitoclax (Zhu et al., 2016). While we cannot determine whether the proliferating cells were previously non-senescent or re-entering the cell cycle after partial reversal of senescence (Beauséjour et al., 2003), our findings suggest that the presence of senescent cells may suppress proliferation within aged endothelial populations. This suppression is likely mediated, at least in part, by paracrine factors such as those found in the SASP, which includes inflammatory cytokines and proteases that damage neighboring cells and disrupt the extracellular matrix (Coppé et al., 2010; Dong et al., 2024). Following senescent cell clearance by Navitoclax or Talabostat, we observed increased population doublings and BrdU incorporation, indicating enhanced proliferative activity. These results support the idea that removing senescent ECs may improve the overall proliferative environment within aged endothelial cultures, regardless of whether this effect is mediated by non-senescent cells or reversal of senescence in previously arrested cells.

In contrast to Navitoclax, treatment with Talabostat led to a reduction in mean telomere length in aged endothelial cells. Telomeres are repetitive nucleotide sequences that cap the ends of chromosomes and protect genomic stability during cell division (De Lange, 2009; Fagagna et al., 2003; Von Zglinicki et al., 2005). Telomere length reflects a cell's proliferative history: highly proliferative cells typically exhibit shorter telomeres due to replication-associated attrition, while quiescent cells retain longer telomeres (Daniali et al., 2013; Maximov et al., 2017). In our study, Talabostat-treated cultures had a lower mean telomere length and a higher frequency of short telomeres compared to the vehicle group, suggesting a scenario in which Talabostat selectively eliminates cells with the shortest and most dysfunctional telomeres, allowing the remaining cells to proliferate and repopulate the culture. This compensatory proliferation might lead to some additional telomere attrition, but, within our experimental timeframe, does not re-establish detectable telomere dysfunction. Whether longer-term proliferation eventually reintroduces TIFs will require future investigation. Our results are in line with our previous findings, showing that endothelial telomere length is reduced following senolytic treatment (Bloom et al., 2023c). Importantly, our TIF assay detects telomere-associated DNA damage by colocalization of 53BP1 with telomeric DNA; therefore, any persistent dysfunctional telomeres would have been captured by our method. The observation of increased cell replication following senescent clearance is supported by telomere length distribution analyses at day 10 and increased BrdU incorporation at day 29. Importantly, despite shorter telomeres, Talabostat-treated cells showed reduced levels of DNA damage and TIFs, indicating that the remaining telomeres were functional. In contrast, Navitoclax increased overall cell proliferation without significantly altering mean telomere length or the frequency of short telomeres. One possibility is that Navitoclax selectively cleared senescent cells, thereby enriching for a population of non-senescent ECs with relatively longer telomeres (Kirkland and Tchkonia, 2017). Alternatively, improved environmental conditions following SASP reduction may have supported cell proliferation without requiring extensive replication and telomere attrition (Coppé et al., 2010). Although telomerase activity was not measured in this study, we cannot exclude the possibility that telomerase contributed to telomere maintenance in Navitoclax-treated cells (Vasa et al., 2000; Shay and Wright, 2019). Overall, these findings suggest that Talabostat and Navitoclax may enhance proliferation through distinct mechanisms, one involving replication-associated telomere shortening and the other through telomere-preserving or telomere-independent pathways.

DNA damage is a key driver of cellular senescence and contributes to endothelial dysfunction with aging (Bloom et al., 2022; Rodier et al., 2009; Bautista-Niño et al., 2016; Kciuk et al., 2023). Telomere shortening or uncapping can activate the DNA damage response (DDR), via p53 and p21, ultimately leading to cell cycle arrest (Liu et al., 2019). In our study, Talabostat treatment reduced both general DNA damage, as indicated by decreased 53BP1 foci, and telomere-specific damage, as shown by reductions in telomere dysfunction-induced foci (TIFs). These changes suggest that Talabostat may improve genomic stability and support regenerative potential in aged endothelial populations. In contrast, Navitoclax did not reduce 53BP1 foci or TIFs, with levels comparable to the vehicle group. Navitoclax induces apoptosis by inhibiting BCL-2 family proteins (Pandravada and Sandler, 2021), but this process is known to elevate mtROS, which can promote oxidative DNA damage (Bloom et al., 2023b; Vorobjeva et al., 2020). Although senescent cells already exhibit increased mtROS (Lafargue et al., 2017), we found that Navitoclax further elevated mtROS levels relative to both vehicle and Talabostat. This may explain its limited impact on DNA damage markers in our model. Our findings align with prior studies showing that BCL-2 inhibition can increase oxidative stress and genomic instability (Bloom et al., 2023b; Robert and Wagner, 2020). Notably, Sharma et al. (2020) reported that Navitoclax impaired bone matrix production in mice, potentially through a similar ROS-driven mechanism (Sharma et al., 2020). While ROS was not directly assessed in that study, our mtROS data offer a plausible mechanistic link. Together, these findings suggest that Talabostat may have a mechanistic advantage over Navitoclax in reducing DNA damage and preserving genomic integrity in aged endothelial cells.

In conclusion, this study demonstrates that both Navitoclax and Talabostat effectively reduce the burden of replicative senescent endothelial cells and are associated with enhanced replicative capacity in aged endothelial cell cultures. These effects were more pronounced with Talabostat, which also reduced markers of DNA damage and telomere dysfunction while lowering mitochondrial ROS. In contrast, Navitoclax treatment increased mtROS and failed to improve genomic integrity, highlighting mechanistic differences between these senolytic agents. While both compounds improved EC proliferation, Talabostat exhibited additional protective effects that may offer advantages for preserving endothelial health. Future studies will be needed to determine whether these benefits are mediated primarily through reductions in SASP signaling or additional paracrine effects. These findings provide mechanistic insight into how distinct senolytics influence aging phenotypes in endothelial cells and support further exploration of Talabostat as a candidate for targeting vascular aging.

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CRediT authorship contribution statement

Hossein Abdeahad: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Denisse G. Moreno:** Validation, Formal analysis, Data curation. **Samuel I. Bloom:** Funding acquisition, Conceptualization. **Hui Huang:** Data curation. **Lisa A. Lesniewski:** Supervision. **Anthony J. Donato:** Supervision.

Originality and exclusivity

This manuscript is original, has not been published previously, and is not under consideration for publication elsewhere. We affirm that the content represents our own work and that proper citations have been provided for all referenced material.

Ethical compliance

The research described in this manuscript complies with all relevant ethical guidelines. No human or animal subjects were involved beyond established cell culture practices.

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Declaration of competing interest

The authors declare no competing interests.

Data availability

All data supporting the findings of this study are included in the manuscript or available from the corresponding author upon reasonable request.

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